

CS 4700/6700 DB Design Project

Fall 2026

Due day: December 4th, Friday

DB design and implementation project can be performed by the following steps:

1. Pick up an application problem of your own.
 2. Entity-Relationship model design.
 3. Convert the ER design to an equivalent Relational DB schema.
 4. Create the relations using MS Access, Oracle, MySQL or some other DBMS.
 5. Insert sample tuples into the relations.
 6. Write some meaningful SQL queries and run them against the sample relations.
 7. Print out the states (i.e., contents) of relations and the results of SQL queries.
 8. Add your own discussion.
- If you recycle an ER model given in a book, then you will lose points because the maximum score for ER modeling is 6 out of total 30 project points, and obviously I cannot give much credit for just copying some existing ER model.
 - Your ER design should include some relationship types, so that some of your SQL queries can involve more than one relation. In general, a relational DB design including more than 4 relations is okay.
 - Try most of the typical SQL commands (described in Chapters 6 and 7) to see how they work.
 - **Your final report should be a single pdf file** including the description of the problem, data storage requirements (i.e., data to be stored and constraints on them), ER design, relation schemas and states (i.e., contents), query descriptions and SQL queries, query outputs, and discussion.
 - The discussion section can include anything that you want to share with other students in our class, such as your comments on the logical DB design, implementation, DBMS functionality and performance, trial and error, future work, etc. You can also include any technical things that can demonstrate how much effort you made on the project.
 - A guideline for using MS Access is given at <http://web1.cs.wright.edu/~schung/cs6700.htm>
 - Grading: Your final report will be graded based on {description of problem 4%, ER design 6%, relational DB schema 5%, SQL queries and results 8%, discussion 7%}